

Exp. 3	Setting Up Hybrid Connectivity Experiment with setting up a secure VPN connection between an on-premises data center and a public cloud provider such as AWS or Azure
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AIM

To implement a Hybrid Cloud environment using Amazon Web Services Free Tier by configuring a Virtual Private Cloud (VPC) with subnets and route tables, and establishing secure connectivity with a local on-premises system using OpenVPN.

OBJECTIVE

- To create a real VPC in AWS Free Tier.
- To configure subnets and route tables.
- To deploy an EC2 instance in cloud.
- To simulate on-premises system using VirtualBox.
- To establish secure VPN connectivity between cloud and local system.
- To understand Hybrid Cloud architecture practically.

SOFTWARE REQUIREMENTS

- AWS Free Tier Account
- VirtualBox
- Ubuntu 20.04 / 22.04 LTS (Local VM + EC2)
- OpenVPN
- Web Browser
- SSH Client (Optional)

THEORY

Hybrid Cloud is a computing model that combines on-premises infrastructure with public cloud services.

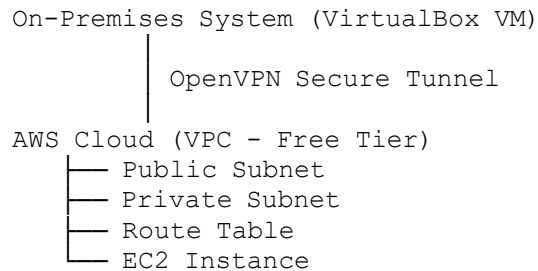
In this experiment:

- AWS provides cloud infrastructure (VPC, EC2, networking)
- Local VirtualBox VM acts as on-premises system
- OpenVPN provides secure encrypted communication

AWS COMPONENTS:

- **VPC (Virtual Private Cloud):** Isolated cloud network
- **Subnet:** Divides VPC into smaller networks
- **Route Table:** Controls traffic flow
- **Internet Gateway:** Enables internet access
- **EC2 Instance:** Virtual machine in cloud

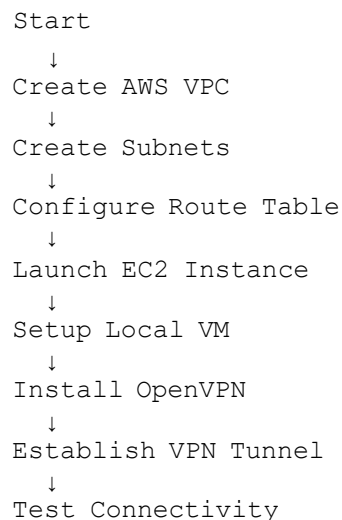
ARCHITECTURE



ALGORITHM

1. Create AWS Free Tier account.
2. Create VPC with CIDR block.
3. Create public and private subnets.
4. Attach Internet Gateway.
5. Configure route tables.
6. Launch EC2 instance.
7. Setup local VM using VirtualBox.
8. Install OpenVPN on both systems.
9. Configure VPN connection.
10. Verify connectivity using ping.

FLOWCHART



PROCEDURE

STEP 1: CREATE VPC IN AWS

1. Login to AWS Console.
2. Go to VPC Dashboard.
3. Click “Create VPC”.
4. Enter:
 - o Name: Hybrid-VPC
 - o CIDR: 10.0.0.0/16
5. Create VPC.

STEP 2: CREATE SUBNETS

- Public Subnet: 10.0.1.0/24
- Private Subnet: 10.0.2.0/24

Associate both with VPC.

STEP 3: CREATE INTERNET GATEWAY

1. Create Internet Gateway (IGW).
2. Attach it to VPC.

STEP 4: CONFIGURE ROUTE TABLE

Add route:

Destination	Target
0.0.0.0/0	Internet Gateway

Associate with Public Subnet

STEP 5: LAUNCH EC2 INSTANCE

1. Go to EC2 → Launch Instance
2. Choose Ubuntu (Free Tier eligible)
3. Select VPC and Public Subnet
4. Enable SSH (port 22)
5. Launch instance

STEP 6: SETUP LOCAL VM (ON-PREMISES)

Using VirtualBox:

- Install Ubuntu Linux
- Assign IP: 192.168.1.10

STEP 7: INSTALL OPENVPN

On both AWS EC2 and Local VM:

```
sudo apt update
sudo apt install openvpn -y
```

STEP 8: CONFIGURE VPN SERVER (AWS EC2)

```
sudo apt install easy-rsa -y
sudo systemctl start openvpn
sudo systemctl enable openvpn
```

STEP 9: CONFIGURE VPN CLIENT (LOCAL VM)

```
sudo openvpn --config client.ovpn
```

STEP 10: VERIFY CONNECTIVITY

From Local VM:

```
ping <EC2-private-IP>
```

From AWS EC2:

```
ping 192.168.1.10
```

COMMANDS USED

```
sudo apt update

sudo apt install openvpn -y

sudo apt install easy-rsa -y

sudo systemctl start openvpn

sudo systemctl enable openvpn

ping <ip-address>
```

OBSERVATIONS

Parameter	Status
VPC Creation	Successful
Subnet Configuration	Successful
Route Table Setup	Successful
EC2 Instance Deployment	Successful
VPN Setup	Successful
Connectivity Test	Successful

EXPECTED OUTPUT

- AWS VPC created successfully
- Public and private subnets configured
- Route tables properly assigned
- EC2 instance running in cloud
- Local VM connected as on-premises system
- Secure VPN tunnel established
- Successful communication between cloud and local system

RESULT

A Hybrid Cloud environment was successfully implemented using AWS Free Tier VPC and a local VirtualBox system. Secure communication between cloud and on-premises systems was established using OpenVPN.